



JAIPUR NATIONAL UNIVERSITY, JAIPUR

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BTEBE203
/ BSC203

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School of Engineering & Technology

B.Tech. II Sem : Biomedical / Biomedical+MBA /

Civil / Electrical / Electronics / Mechanical / Chemical /

Computer Science / CSE (AIML) / CSE (Cyber Security) /

II SEMESTER END EXAMINATION: MAY 2022

ENGINEERING MATHEMATICS - II

Time: 3 Hrs.

Max. Marks: 70

INSTRUCTIONS

All questions are compulsory.

UNIT-I

Q.1 (a) Find the rank of a matrix:

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$$\begin{bmatrix} 2 & 10 & 9 & 7 & 8 \\ 1 & 2 & 4 & -1 & 1 \\ 3 & 3 & -2 & 1 & -1 \\ 2 & 6 & 9 & 12 & 1 \end{bmatrix}$$

(b) Find the value of λ_1 & λ_2 such that

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$$\begin{cases} x + 2\lambda_1 y + z = \lambda_2 \\ x + 2y + \lambda_1 z = 1 \end{cases}$$

$$\lambda_1 x + 2y + z = 1$$

has (i) Unique solution

(ii) No solution

OR

(a) Find the eigen value & eigen vector of the matrix:

$$A = \begin{bmatrix} 1 & -1 & -1 \\ 2 & -3 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

(b) Solve the system of equation:

$$x + y + z = 0$$

$$2x + 5y + 7z = 0$$

$$2x - 5y + 3z = 0$$

UNIT-II

Q.2 (a) Find the fourier series of $x \cos x$ in the interval $(-\pi, \pi)$

(b) Find the Fourier series of $f(x) = x - x^2$ in the interval $(-\pi, \pi)$

OR

(a) Find the fourier series constant term and coefficient of first sine term of:

x	0	1	2	3	4
y	8	9	3	6	9

(b) Find the half range sine series of:

$$1(-6-1) + 1(4) - 1(2) = -7 + 4 - 2 = -5$$

$$f(x) = x^3, x \in (0, 2)$$

UNIT-III

a) Solve: $(x^2 + y^2 + 2x) dx + 2y dy = 0$

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b) Solve: $(D^2 + 2D + 1)y = e^x + x^2 - \sin x$

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OR

(a) Solve: $(D^2 + 2D + 1)y = x \sin x$

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(b) Solve: $(D^2 - 2D + 1)y = x e^x \sin x$

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UNIT-IV

Q.4

Solve: $(1+x)^2 \frac{d^2y}{dx^2} + (1+x) \frac{dy}{dx} + y = 4 \cos \log(1+x)$

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$x+1 = e^z$
 $z = \log(x+1)$

$D(D-1)y + Dy = 4 \cos z$ OR

Solve: $x^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} - 3y = x^2 \log x$

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$x = e^z$
 $z = \log x$

UNIT-V

Q.5 (a) Solve: $x^2(y-z)p + y^2(z-x)q = z^2(x-y)$

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(b) Solve: $z(x+y)p + z(x-y)q = x^2 + y^2$

7

OR

(a) Solve: $p = (qy + z)^2$

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